

DLA CT-2 Important Questions

1. A hospital is developing an automated system to detect pneumonia from chest X-ray images. The system must accurately identify infected regions and assist doctors in diagnosis. Explain how a Convolutional Neural Network (CNN) can be designed for this task with various architecture components, feature extraction process, training strategy and evaluation metrics. Explain the challenges in medical image classification and how CNN overcomes them.
2. A hospital is developing a smart monitoring system to track ICU patients using continuous data such as heart rate, blood pressure, and oxygen levels over time. The goal is to predict critical conditions early. To handle this sequential data efficiently, a GRU-based model is proposed. Explain how a GRU model can be designed for this task, including its architecture and gating mechanisms. Discuss how it processes time-series data, captures dependencies, and how the model is trained and evaluated. Also explain the challenges in sequential medical data and how GRU helps address them.
3. A hospital is developing a system to monitor ICU patients using continuous data such as heart rate, blood pressure, and oxygen levels recorded over time. The goal is to predict critical conditions like cardiac arrest in advance. Since the data is sequential, a Recurrent Neural Network (RNN) is proposed. Explain how an RNN-based model can be designed for this task. Discuss how the model learns from time-based data, handles long-term dependencies, and how it is trained and evaluated. Also explain the challenges in processing sequential medical data.
4. An IoT-based healthcare system continuously monitors patient vitals (heart rate, ECG, oxygen levels) to predict critical health events in advance. Explain how a Long Short-Term Memory (LSTM) network can be used for this task. Include: Internal structure (input, forget, output gates), Handling long-term dependencies, Advantages over traditional RNN, and Application workflow for real-time prediction.
5. A 5x5 grayscale image and a 3x3 filter is given. Elaborate on how the convolution operation is performed and calculate the size of the output feature map. Also, explain the role of stride and padding in determining the output size with the examples.
6. The task assigned to you is developing an AI system to classify short videoclips into categories such as "running," "dancing," or "playing guitar." Each clip is a sequence of 30 image frames (grayscale). Describe how you would build a model that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) to solve this problem. What part of the model would use CNN and why?
7. How would LSTM be used after CNN? Illustrate the input shape and data preprocessing steps. What is the benefit of combining CNN with LSTM instead of using either one alone?
8. Describe in detail how the auto encoders and decoders operate in Transformer models.